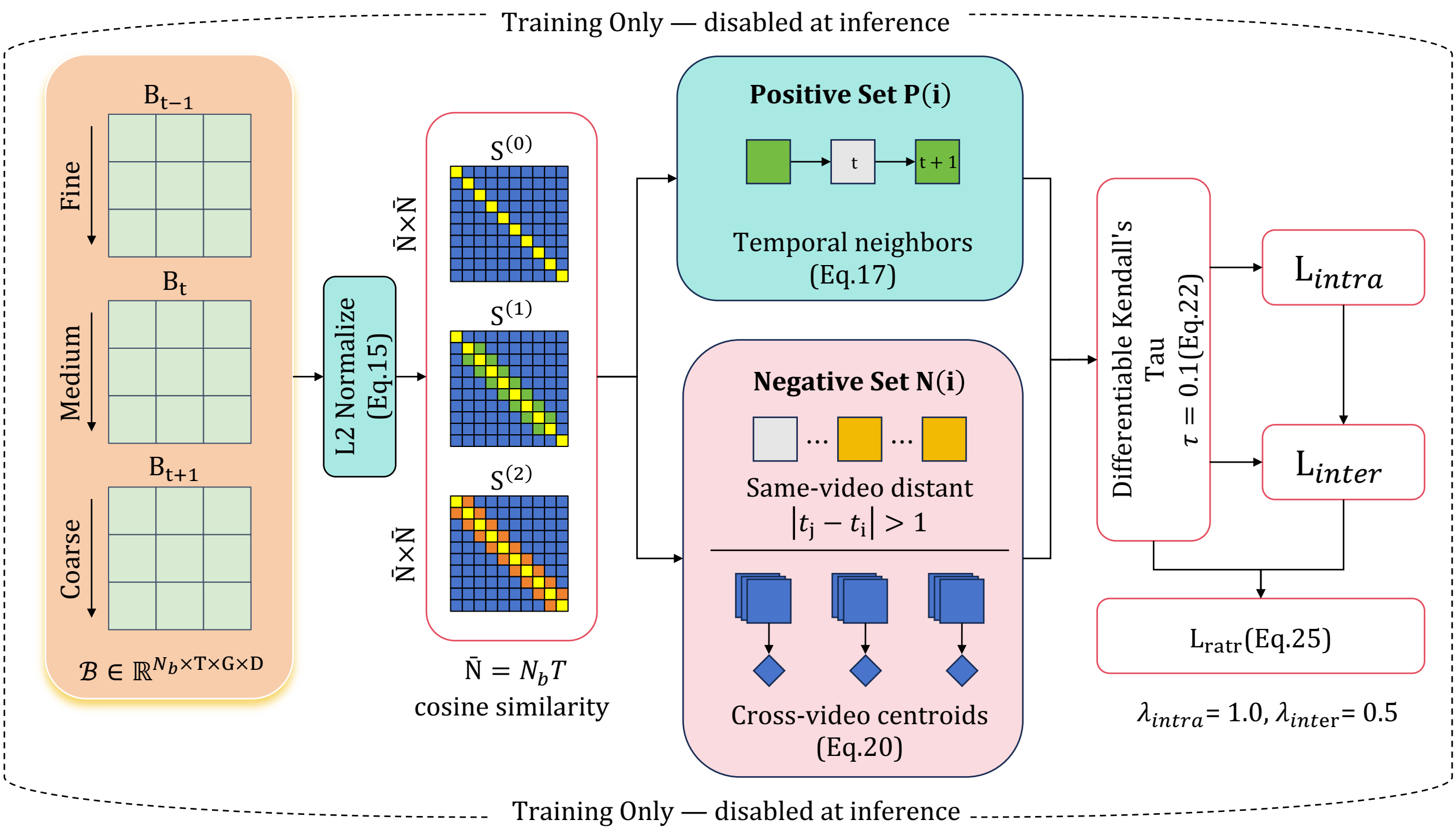

Training Only — disabled at inference
B_{t-1}
Fine
B_t
Medium
B_{t+1}
Coarse
$\mathcal{B} \in \mathbb{R}^{N_b \times T \times G \times D}$
L2 Normalize (Eq.15)
$\bar{N} \times \bar{N}$
$S^{(0)}$
$S^{(1)}$
$S^{(2)}$
$\bar{N} \times \bar{N}$
$\bar{N} = N_b T$
cosine similarity
Positive Set P(i)
t
t + 1
Temporal neighbors (Eq.17)
Negative Set N(i)
Same-video distant
$|t_j - t_i| > 1$
Cross-video centroids (Eq.20)
Differentiable Kendall's Tau
$\tau = 0.1$(Eq.22)
L_{intra}
L_{inter}
L_{ratr}(Eq.25)
$\lambda_{intra} = 1.0, \lambda_{inter} = 0.5$
Training Only — disabled at inference